



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (03/02/2025 – 09/02/2025)

WEEK 3

PROJECT NAME

CHANGE DETECTION

PROJECT DESCRIPTION :

This project aims to detect changes between satellite images captured at different times using Deep Learning and Computer Vision techniques.

GROUP MEMBER :

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GROUP ID :

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GROUP GUIDE :

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COLLEGE NAME :

GATI SHAKTI VISHWAVIDYALAYA

1/06/2026	• Sir, gave us images of different time stamps for change detection, used CV for this and worked with .png images.
2/06/2026	• Performed pixel wise rastering on the images- single band(GRAYSCALE) , RGB. Used absolute difference formula for single band and absolute of square root of bands difference formula for RGB.
3/06/2026	• Downloaded google earth pro and took screenshot of BISAG region. • Understood the conversion of .png images to .tiff.
4/06/2026	• Georeferencing the .png images, downloaded google satellite plugin in QGIS for this and matching the coordinates with the png images for georeferencing.
5/06/2026	• Step towards creating website that can perform this task by just uplading the tif images. • Used antigravity and stitch for this.
6/06/2026	• Generated initial change maps highlighting areas of significant variation between years. Analyzed the effectiveness of different threshold values for separating major and minor changes.
7/06/2026	Holiday (Sunday)

Next Week Plan	Planning to work with model, and exploring more ways through which accuracy can be improved.
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REFERENCE:

- [https://docs.qgis.org/3.44/en/docs/user_manual/working_with_vector/editing_geome try_attributes.html](https://docs.qgis.org/3.44/en/docs/user_manual/working_with_vector/editing_geometry_attributes.html)
- <https://www.earthdata.nasa.gov/learn/data-recipes/detect-environmental-change-using-sar-data-qgis>

SCREEN SHOTS:





